Q1. Which one of the following statements is NOT correct about the B+ treedata structure used for creating an index of a relational database table?

(a) B+ tree is a height-balanced tree

(b) Non-leaf nodes have pointers to data records

(c) Key values in each node are kept in sorted order

(d) Each leaf node has a pointer to the next leaf node

Q2. In a B+ tree, if the search-key value is 8 bytes long, the block size is 512bytes and the block pointer is 2 bytes, then the maximum order of the B+ tree is _____

(a) 52

(b) 54

(c) 56

(d) 57

Q3. B+ trees are considered BALANCED because

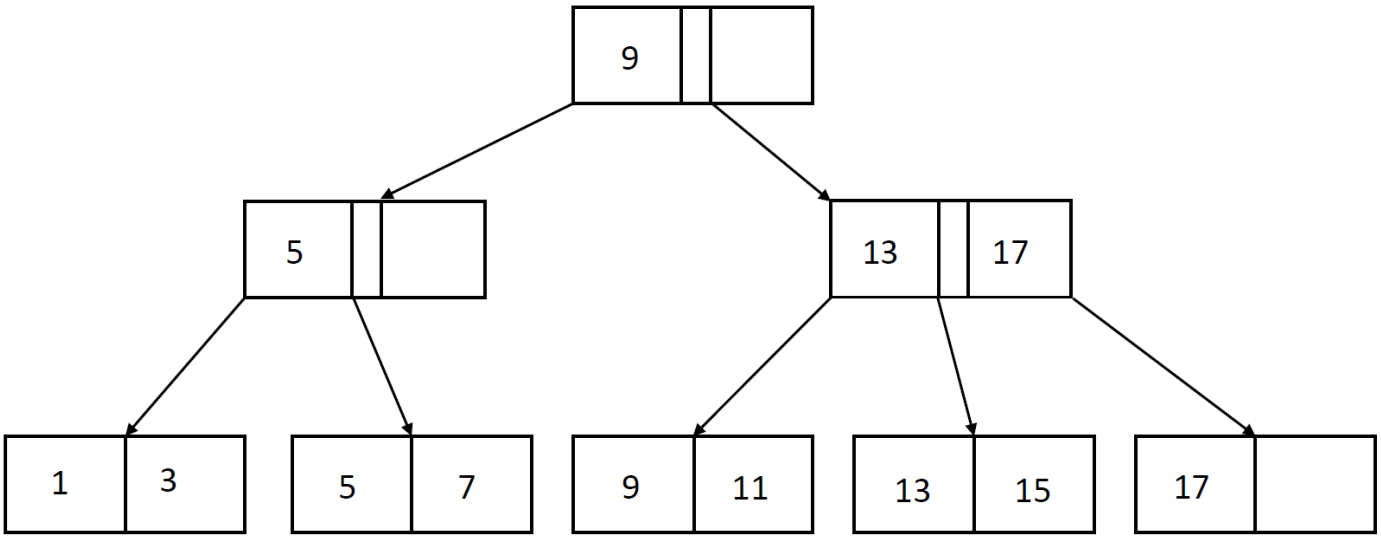
(a) the length of the paths from the root to all leaf nodes are all equal.

(b) the length of the paths from the root to all leaf nodes differ from each otherby at most 1.

(c) the number of children of any two non-leaf sibling nodes differ by at most 1.

(d) the number of records in any two leaf nodes differ by at most 1.

Q4. With reference to the B+ tree index of order 1 shown below, the minimumnumber of nodes (including the root node) that must be fetched in order to satisfy the following query: "Get all records with a search key greater than or equal to 7 and less than 15" is _____



(a) 4
(b) 5
(c) 6
(d) 7

Q5. Consider a B+ tree in which the maximum number of keys in a node is 5. What is the minimum number of keys in any non-root node?

(a) 1
(b) 2
(c) 3
(d) 4

Q6. The order of a leaf node in a tree B+ is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is 1K bytes, the data record pointer is 7 bytes long, the value field is 9 bytes long and a block pointer is 6 bytes long. What is the order of the leaf node?

(a) 63
(b) 64
(c) 67
(d) 68

Q7. A data file consisting of 1,50,000 student records is stored on a hard disk with block size of 4096 bytes. The data file is sorted on the primary key RollNo. The size of a record pointer for this disk is 7 bytes. Each student record has a candidate key attribute called A Num of size 12 bytes. Suppose an index file with records consisting of two fields, A Num value and the record pointer the corresponding student record, is built and stored on the same disk. Assume that the records of data file and index file are not split across disk blocks. The number of blocks in the index file is_____

(a) 698
(b) 898
(c) 899
(d) 4096

Q8. Calculate the order of leaf (pleaf) and non-leaf (p) nodes of a B+ tree based on the information given below:

Search key field = 12 bytes
Record pointer = 10 bytes
Block pointer = 8 bytes
Block size = 1 KB

Which of the following option is true?

(a) pleaf = 51 & p = 46
(b) pleaf = 47 & p = 52
(c) pleaf = 46 & p = 50
(d) pleaf = 52 & p = 47

Q9. An index is clustered, if

(a) it is on a set of fields that form a candidate key
(b) it is on a set of fields that include the primary key
(c) the data records of the file are organized in the same order as the data entries of the index
(d) the data records of the file are organized not in the same order as the data entries of the index.

Q10. A clustering index is defined on the fields which are of type

(a) non-key and ordering
(b) non-key and non-ordering
(c) key and ordering
(d) key and non-ordering

Ans:

1. B
2. A
3. A
4. B
5. B
6. A
7. A
8. C
9. C
10. A

